03.04: packages depens ✓

(torch 2.5.1 has built-in CUDNN 9.1.0)

test for ✓ | SVD, eigdecomp in numpy / scipy

contraction in opt_einsum

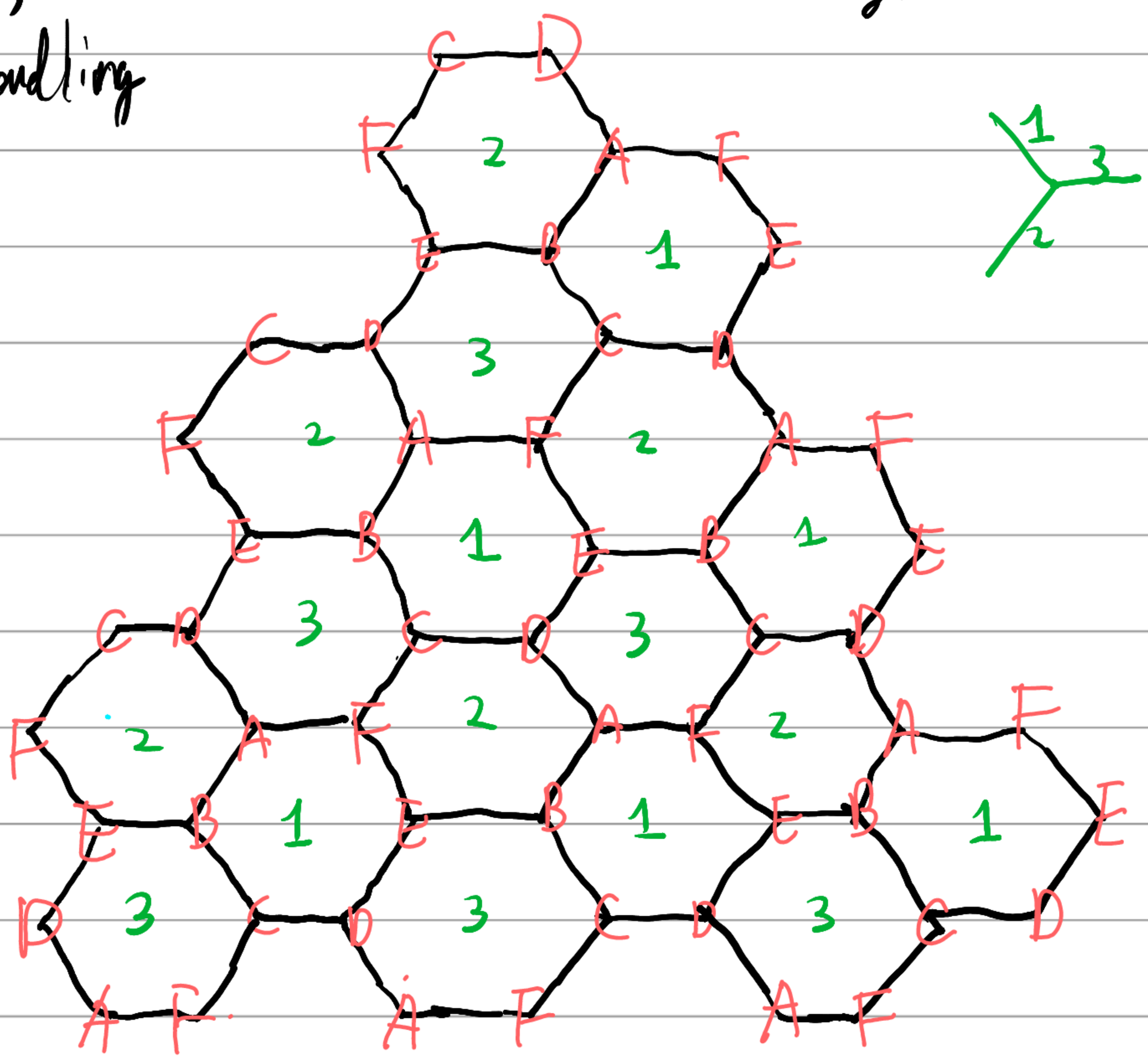
convection, GPU

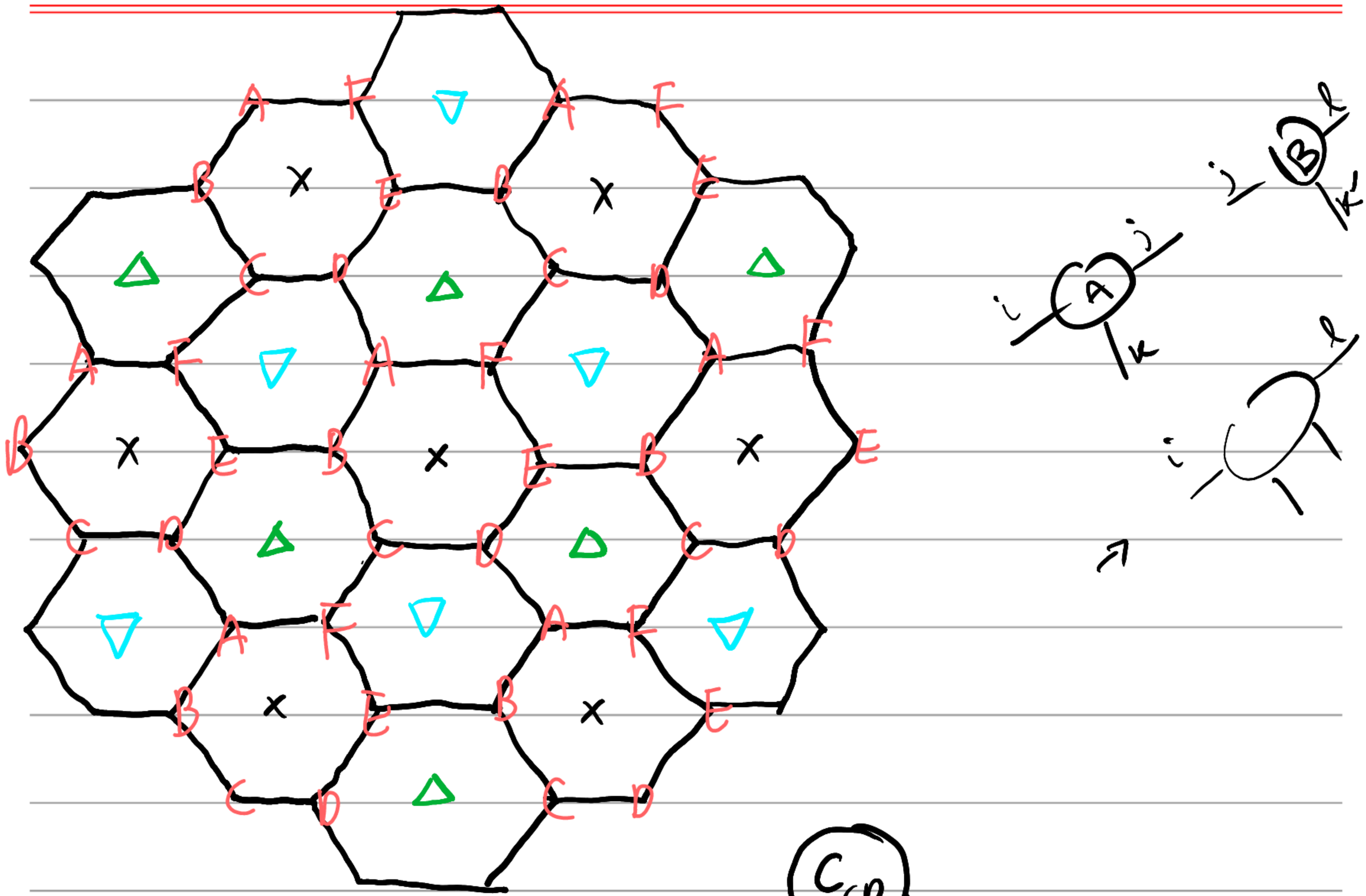
autograd { einsum, SVD, eigh } for torch

truncate SVD !!!

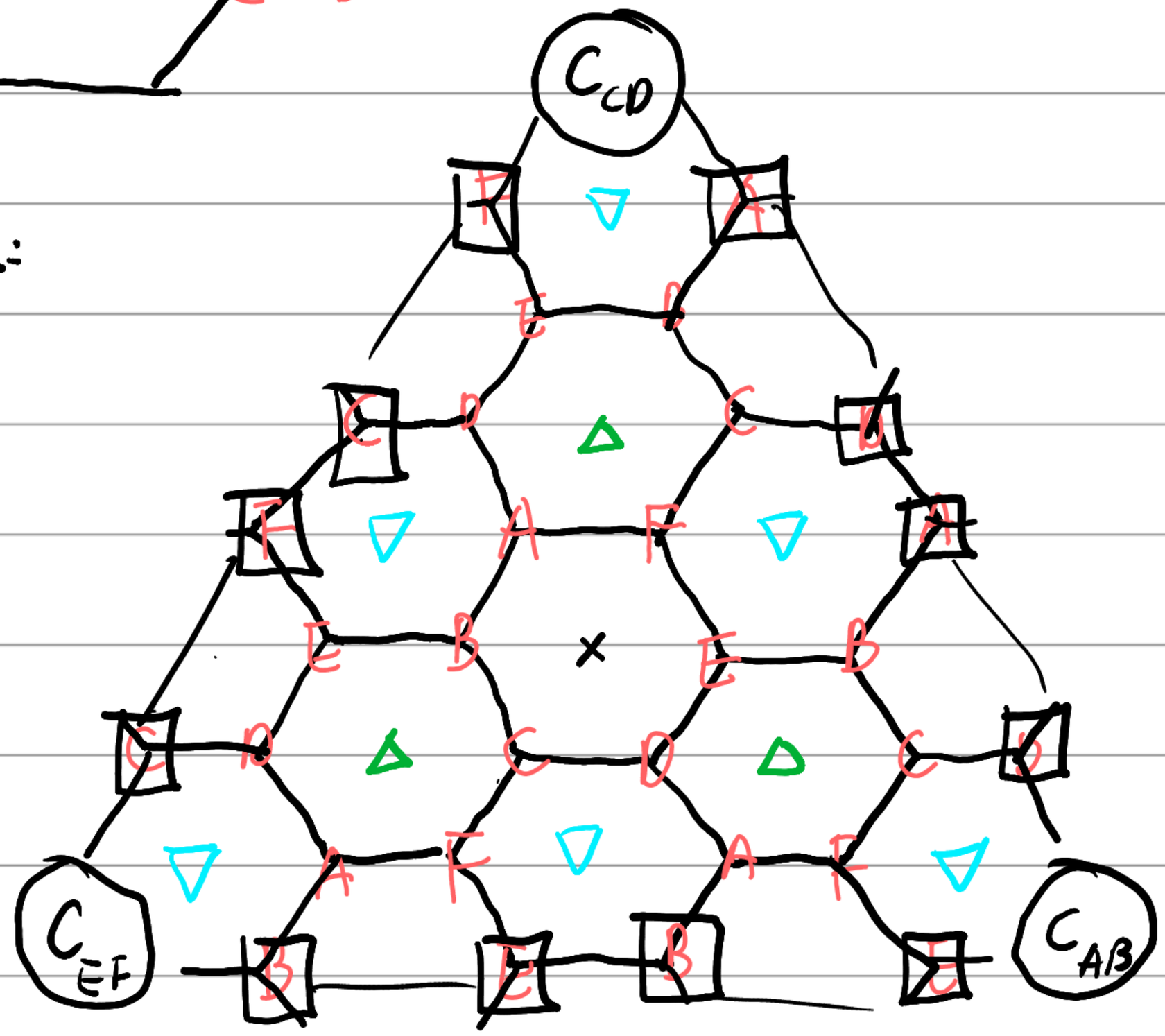
03.05

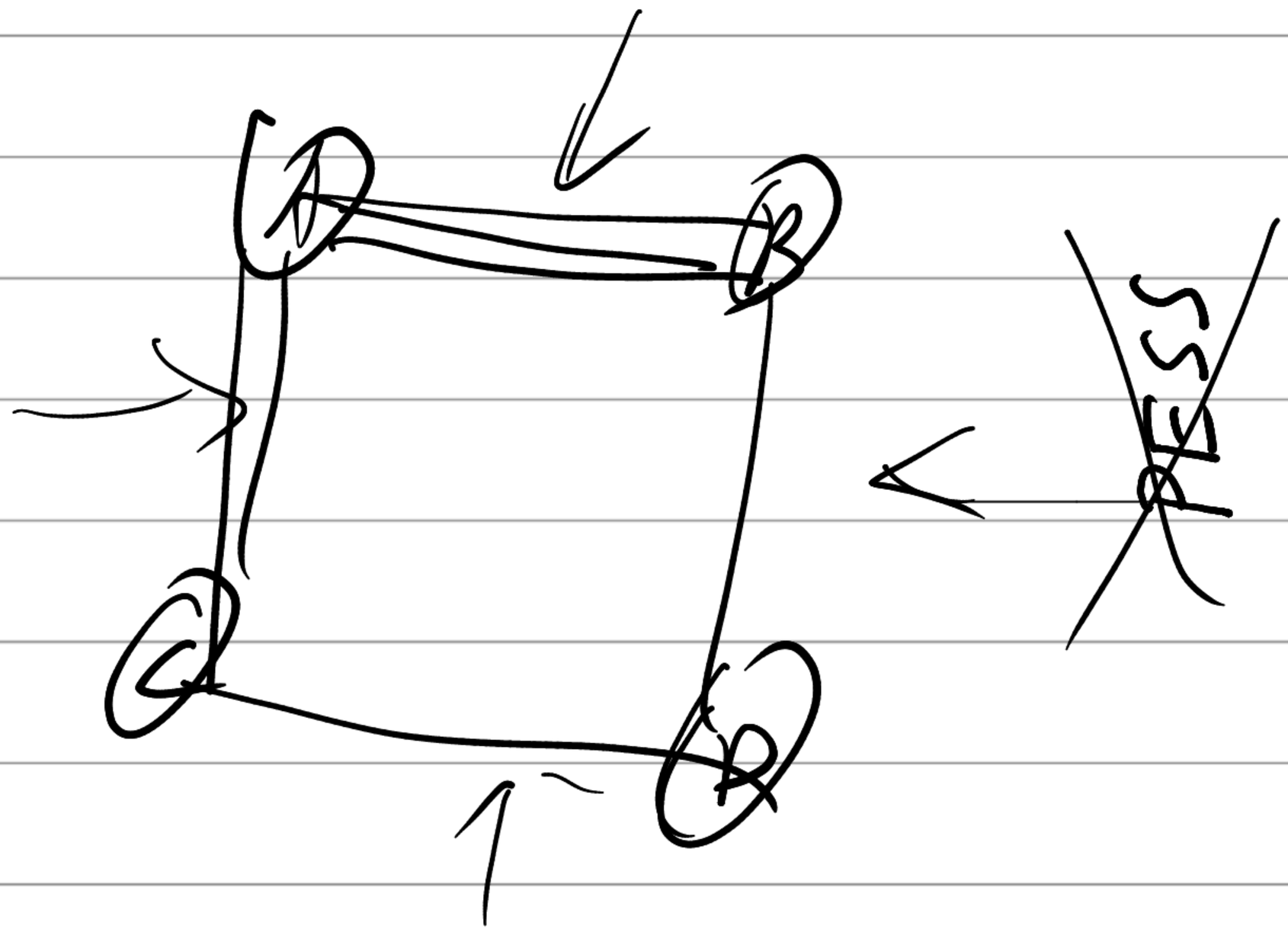
index handling





for center $\boxed{x}$:



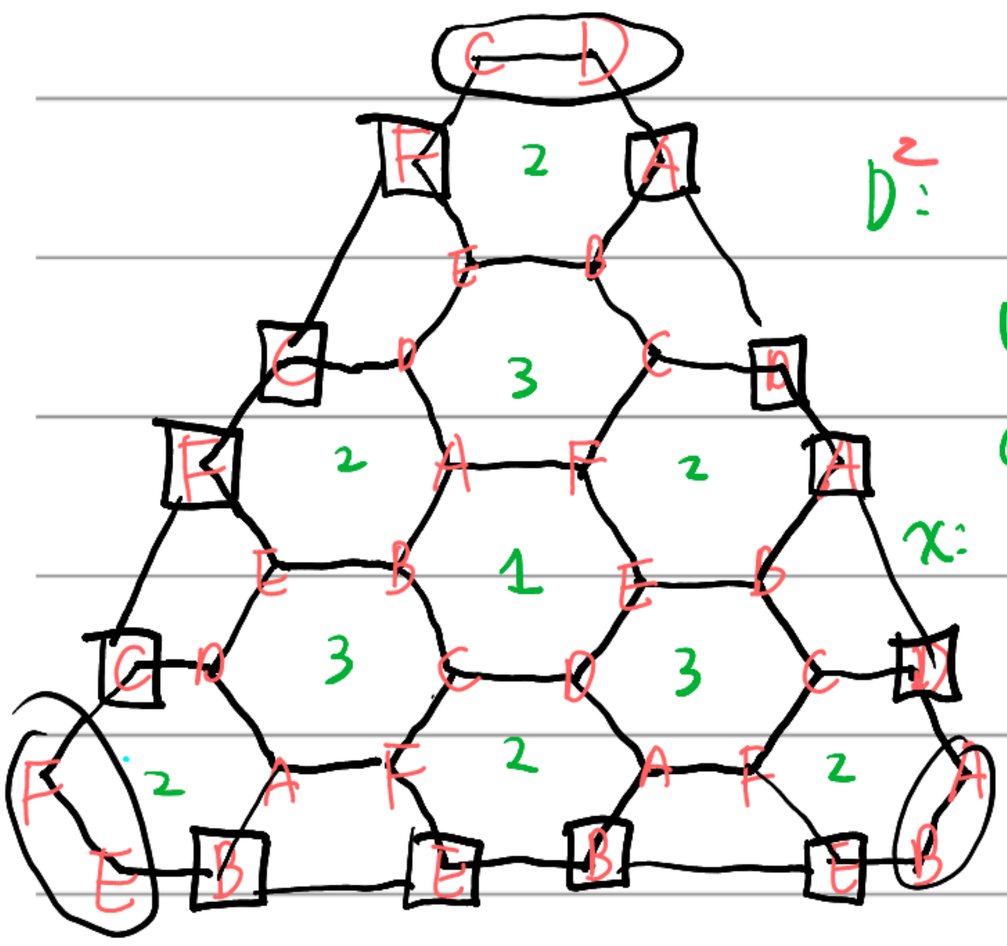


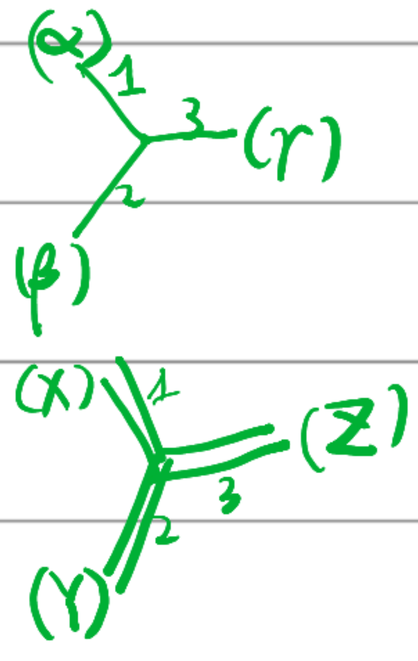
final Σ: $\sim 10^{-5 \sim -6}$

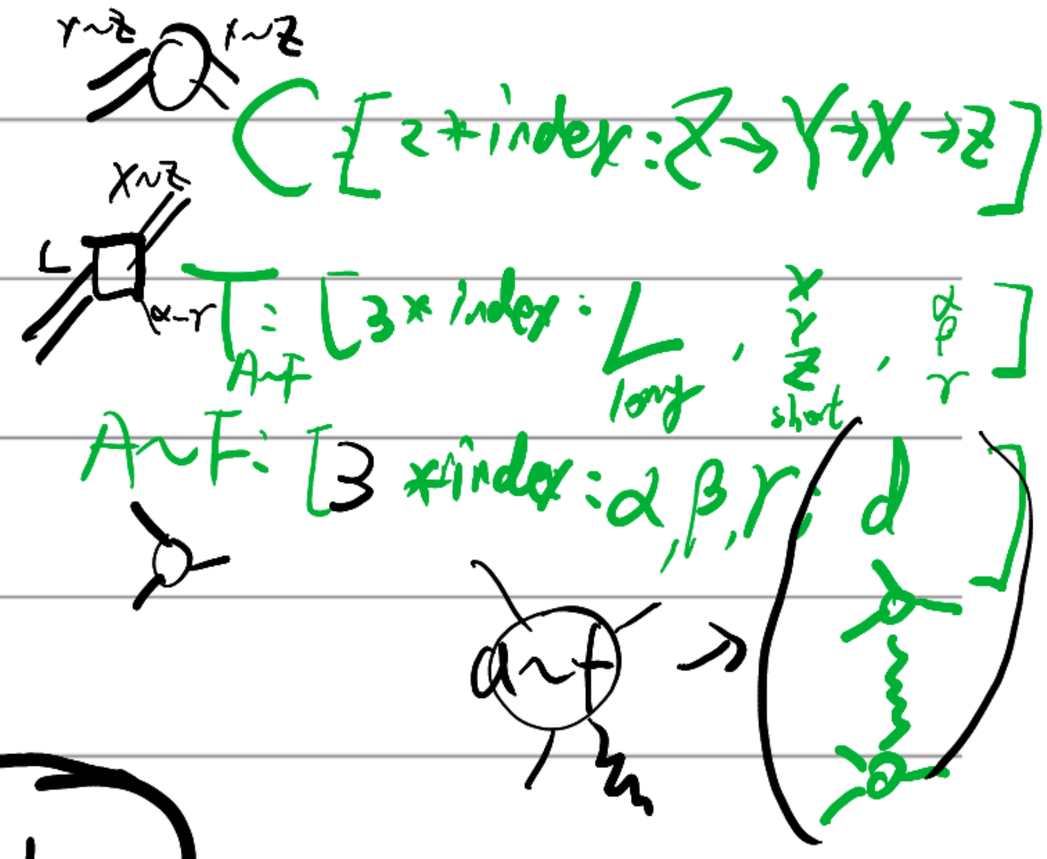
machine precision: float 32

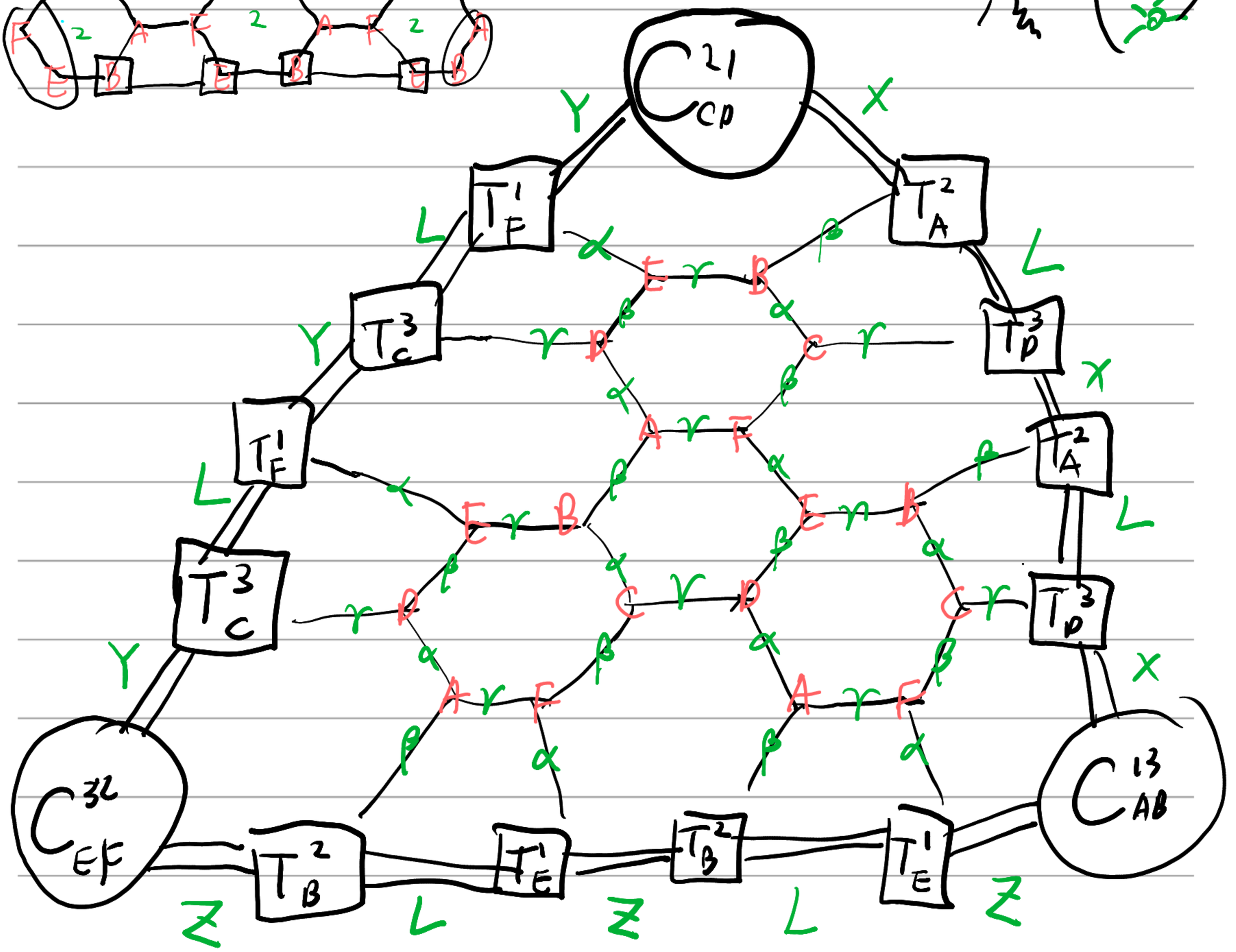
$\therefore$ random SVD 能达 $10^{-9} \sim 10^{-10}$

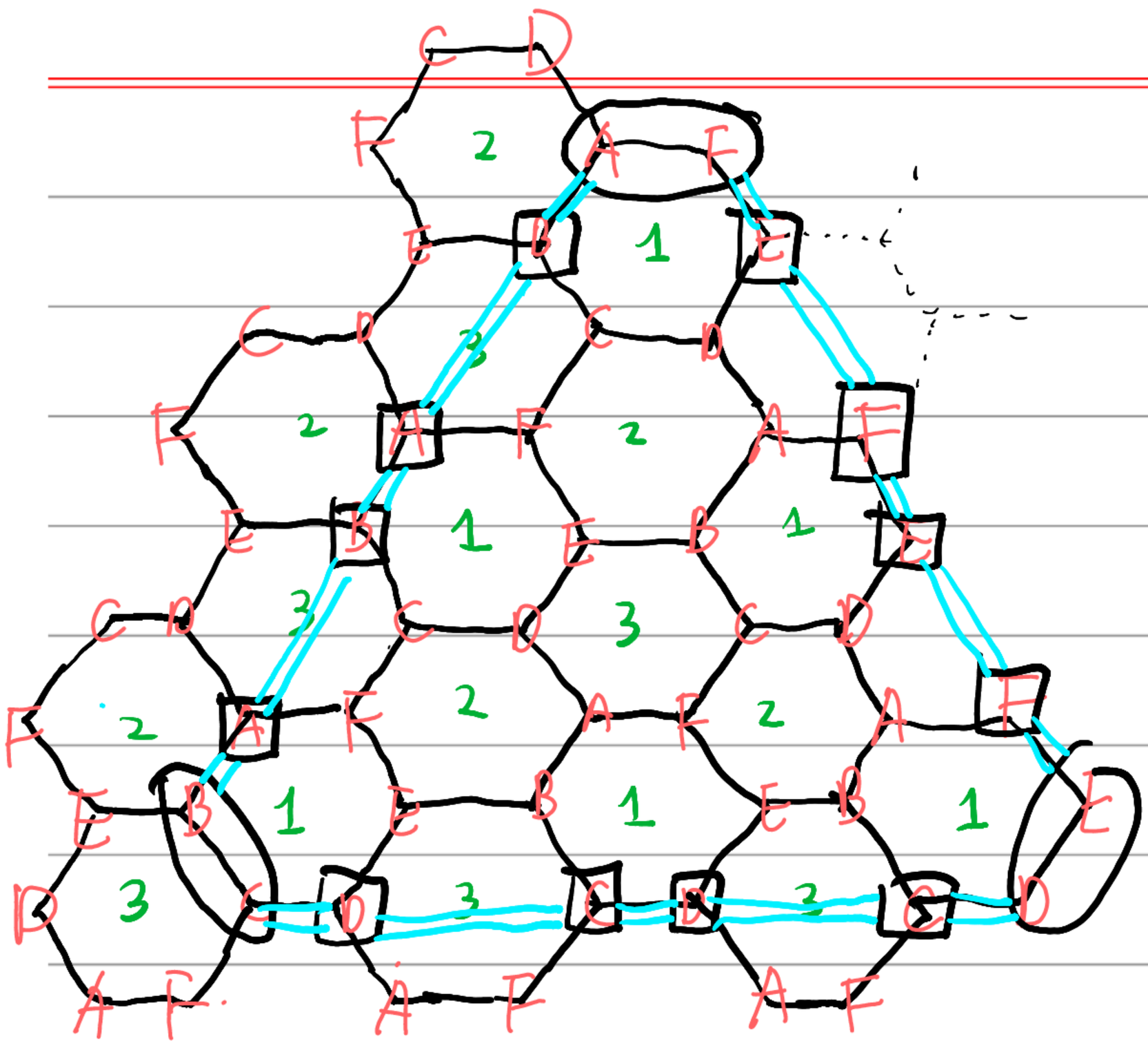
后面test, compared for 正常SVD
for CIMRG

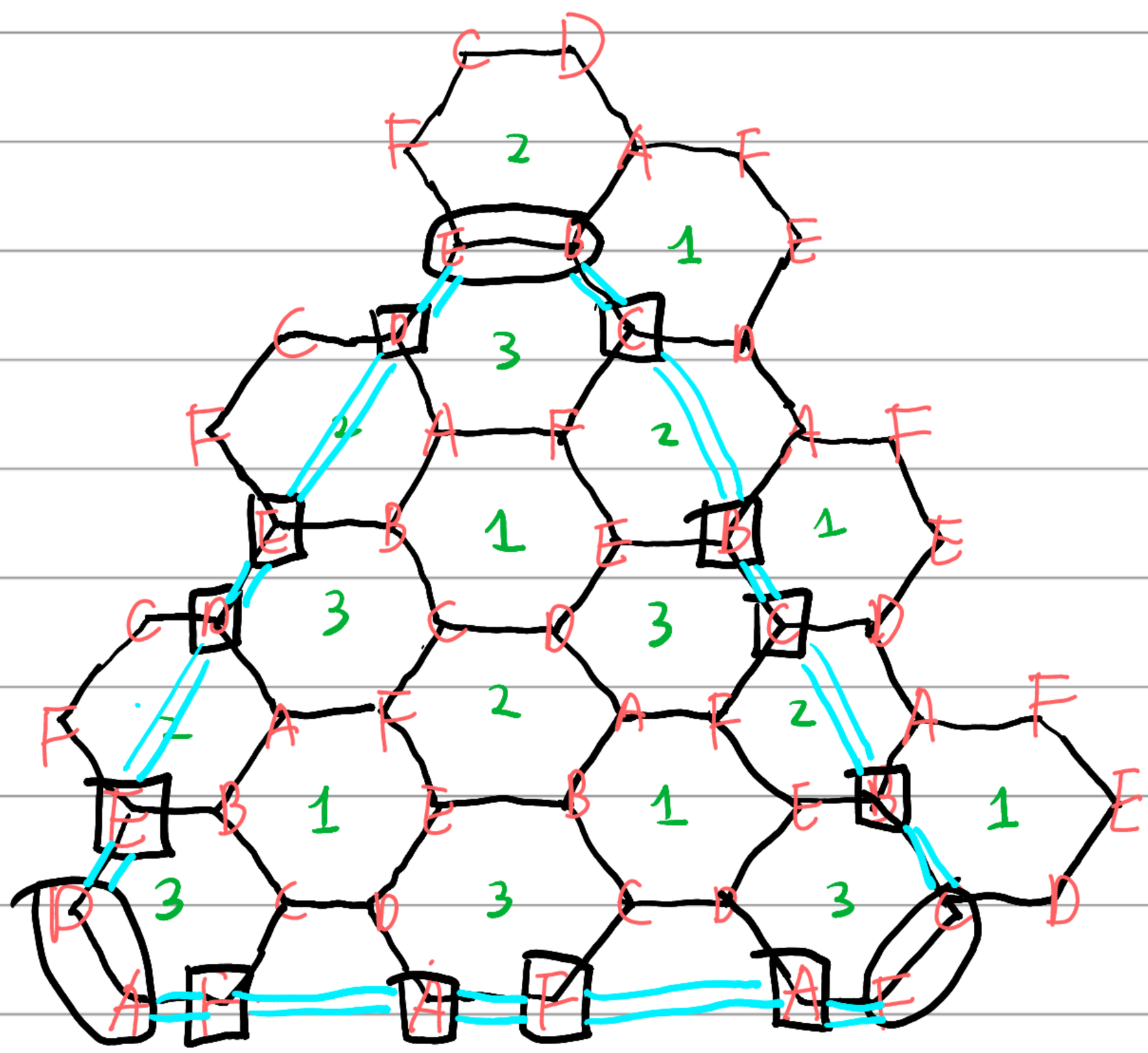



D:
x:


C [2*index: Z→Y→X→Z]
T: [3*index: L, X, α]
A~F: [3*index: α, β, γ]


C CD
C AB
C EF
T F
T A
T C
T D
T B
T E





C^{21}: CD → EB, CD → AF, EB → AF [YX] $\quad T^1 T^2$ [N, Z, αβ]

C^{32}: EF → AD, EF → CB, AD → CB [ZY] $\quad T^2 T^3$ [L, X, βγ]

C^{13}: AB → CF, AB → ED, CF → ED [XZ] $\quad T^3 T^1$ [M, Y, γα]

$C(X \times X)$ [Z→Y→X→Z *2]

$T(X \times X \times D^2)$ [Longer X, M Y, N Z, αβγ]

$A \sim F(D^2 \times D^2 \times D^2)$ [α, β, γ]

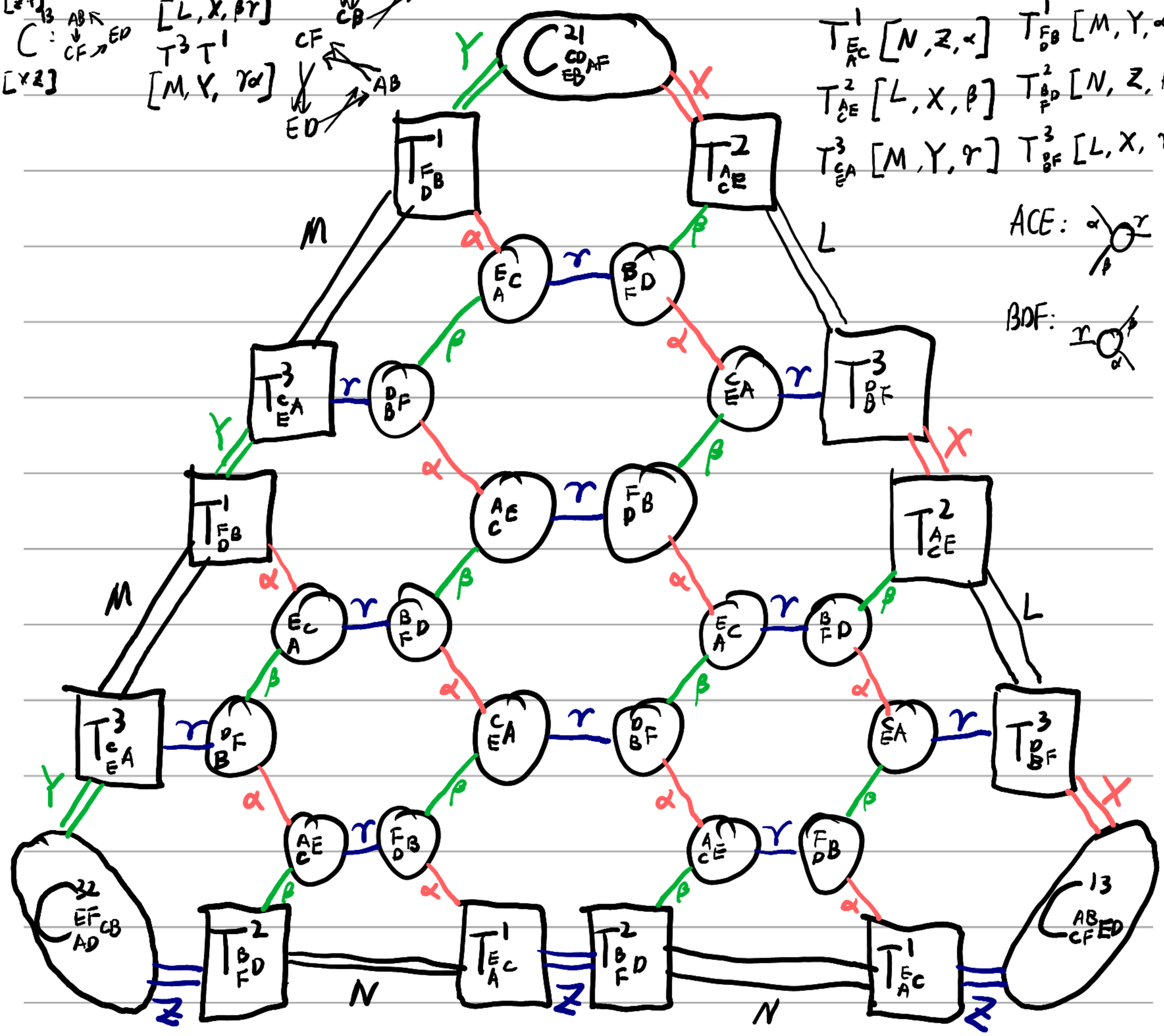
T^1_{EC} [N, Z, α] $\quad T^1_{FB}$ [M, Y, α]

T^2_{AE} [L, X, β] $\quad T^2_{BD}$ [N, Z, β]

T^3_{CA} [M, Y, γ] $\quad T^3_{BF}$ [L, X, γ]

ACE: α γ β

BDF: γ β α

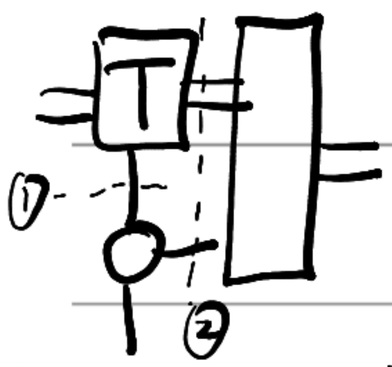


C^{21}: CD → EB, CD → AF, EB → AF $\quad T^1_{EC} \to T^2_{BD} \to T^1_{EC} \to T^2_{BD} \to T^1_{EC} \to T^2_{BD} \to T^1_{EC}$

C^{32}: EF → AD, EF → CB, AD → CB $\quad T^2_{AE} \to T^3_{DF} \to T^2_{AE} \to T^3_{DF} \to T^2_{AE} \to T^3_{DF} \to T^2_{AE}$

C^{13}: AB → CF, AB → ED, CF → ED $\quad T^3_{CA} \to T^1_{FB} \to T^3_{CA} \to T^1_{FB} \to T^3_{CA} \to T^1_{FB} \to T^3_{CA}$

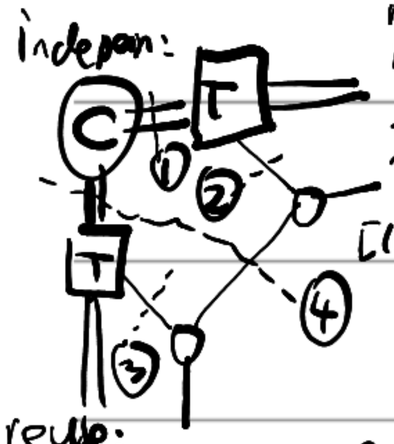
① center $B^{A\,F}_{C\,D\,E}$



max int. memory: $\chi^2(D^2)^2$

FLOPs: $\chi^3(D^2)^2 + \chi^2(D^2)^3$

indepen:



max int. memory: $\chi^2(D^2)^2$

FLOPs: $\chi^2(D^2)^4 + \dots$

[(0,2), (0,3), (1,2), (0,1)]

reuse:

$\chi^3(D^2)^3 + \dots$

[(3,4), (0,1)+3]

$C^{21}_{CD}, T^1_F, T^2_A, E, B$ $\xrightarrow{\text{MANUAL!!}}$ $C^{21\,(big)}_{EB}$

$YX, M\gamma\alpha, L\chi\beta, \alpha m\gamma, \ell\beta\gamma$

$M\,m\,L\,\ell \xrightarrow{\text{reshape}} (M*m, L*\ell)$

exact SVD

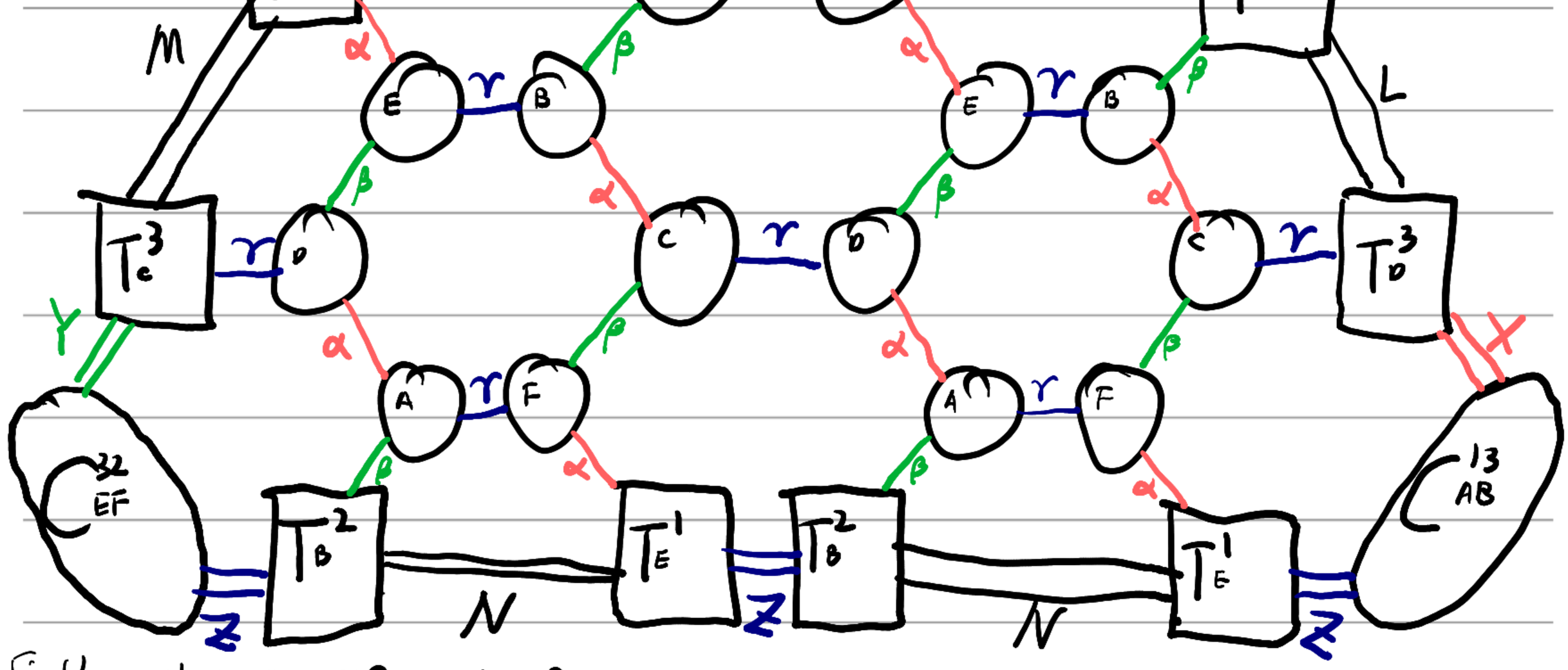
svd_lowrank

⚠ REUSE! $U^3_C, C^{21}_{EB}, U^{3\dagger}_D$

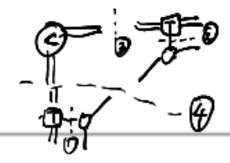
$M*m\,Y,\ YX,\ X\,L*\ell$

$T^1_F, E, U^{1\dagger\,(reshaped)}_F \rightarrow T^3_E$

$O\gamma\alpha, \alpha\beta\gamma, MO\beta$ $\quad M\gamma\gamma$



? Reuse



could be more FLOPs saving in total? (because of $6*$ [diagram] ?)

$\Rightarrow \begin{cases} \chi > D^2: \text{independent} \\ \chi \le D^2: \text{reuse} \end{cases}$

Firstly, all "indepen" & exact SVD.

$C^{21}_{CD}, T^1_F, T^2_A, E, B$ $\xrightarrow{[(0,2),(0,3),(1,2),(0,1)]}$ $C^{21\,(big)}_{EB}$ $\xrightarrow{\text{reshape}}$ $C^{21\,(mat)}_{EB}$ $\xrightarrow[[:\chi]]{\text{SVD}}$ $U^3_C, C^{21}_{EB}, U^{3\dagger}_D$ $\quad T^1_F, E, U^{1\dagger\,(reshape)}_F \rightarrow T^3_E$ $[(0,1),(0,1)]$

$YX, M\gamma\alpha, L\chi\beta, \alpha m\gamma, \ell\beta\gamma$ $\quad M\,m\,L\,\ell$ $\quad M*m, L*\ell$ $\quad M*m, Y\ \ YX\ \ X, L*\ell$ $\quad O\gamma\alpha, \alpha\beta\gamma, MO\beta \rightarrow M\gamma\gamma$

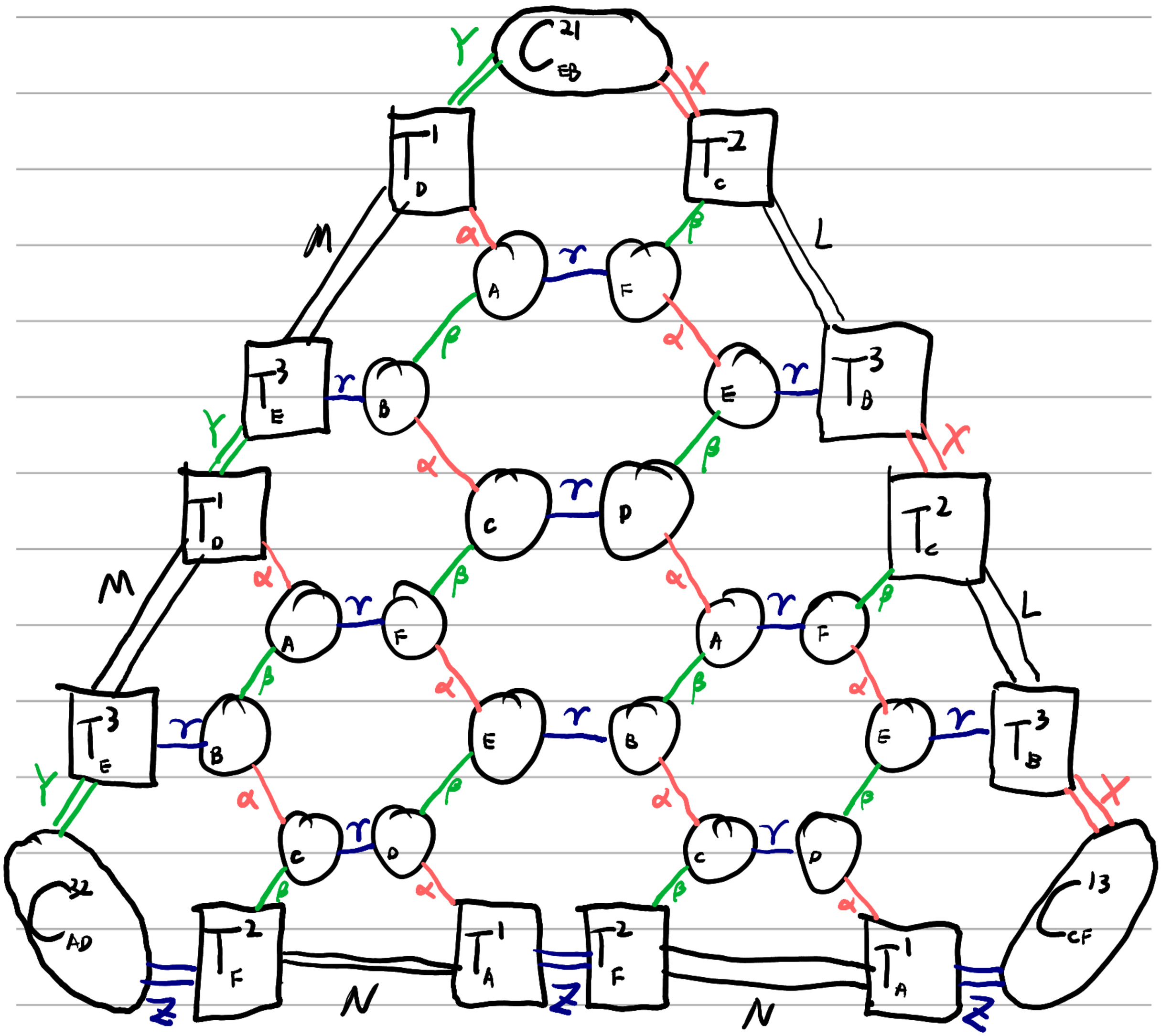
$C^{32}_{EF}\ T^2_B\ T^3_C\ A\ D \rightarrow C^{32\,(big)\rightarrow(mat)}_{AD}$ $\xrightarrow{\text{SVD}}$ $U^1_E, C^{32}_{AD}, U^{1\dagger}_F$ $\quad T^2_A, B, U^2_A \rightarrow T^3_B$ $\quad T^2_B\ A\ U^{2\dagger}_B \rightarrow T^1_A$

$ZY, NZ\beta, M\gamma\gamma, \alpha\beta n\ \alpha m\gamma$ $\quad N*n, M*m$ $\quad Nn, Z\ \ ZY\ \ Y, Mm$ $\quad O\gamma\gamma\ \alpha\beta\gamma\ \ MO\beta$ $\quad O\chi\beta\ \alpha\beta\gamma, LO\alpha$ $\quad L\chi\gamma\ \ O\chi\beta\ \alpha\beta\gamma\ \ NO\gamma\ \ NZ\alpha$

$C^{13}_{AB}\ T^3_D\ T^1_E\ C\ F \rightarrow C^{13\,(big)\rightarrow(mat)}_{CF}$ $\xrightarrow{\text{SVD}}$ $U^2_A, C^{13}_{CF}, U^{2\dagger}_B$ $\quad T^3_D\ C\ U^{3\dagger}_D \rightarrow T^2_C$ $\quad T^1_E\ F\ U^1_E \rightarrow T^2_F$

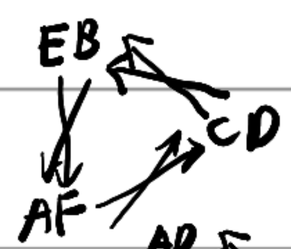
$XZ\ \ L\chi\gamma\ \ NZ\alpha\ \ \ell\beta\gamma\ \ \alpha\beta n$ $\quad L\ell, Nn$ $\quad L\ell, X\ \ XZ\ \ Z, Nn$ $\quad O\chi\gamma\ \ \alpha\beta\gamma$ $\quad LO\alpha\ \ L\chi\beta$ $\quad O\chi\alpha\ \alpha\beta\gamma\ \ NO\gamma\ \ NZ\beta$

② center $\begin{matrix} & C & D & \\ F & & & A \\ & E & B & \end{matrix}$



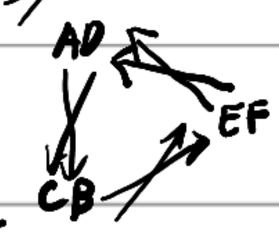
C^{21}: CD → EB → AF [YX]

$T^1 T^2$ [N, Z, αβ]



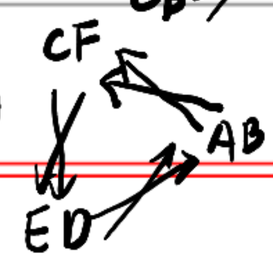
C^{32}: EF → AD → CB [ZY]

$T^2 T^3$ [L, X, βγ]

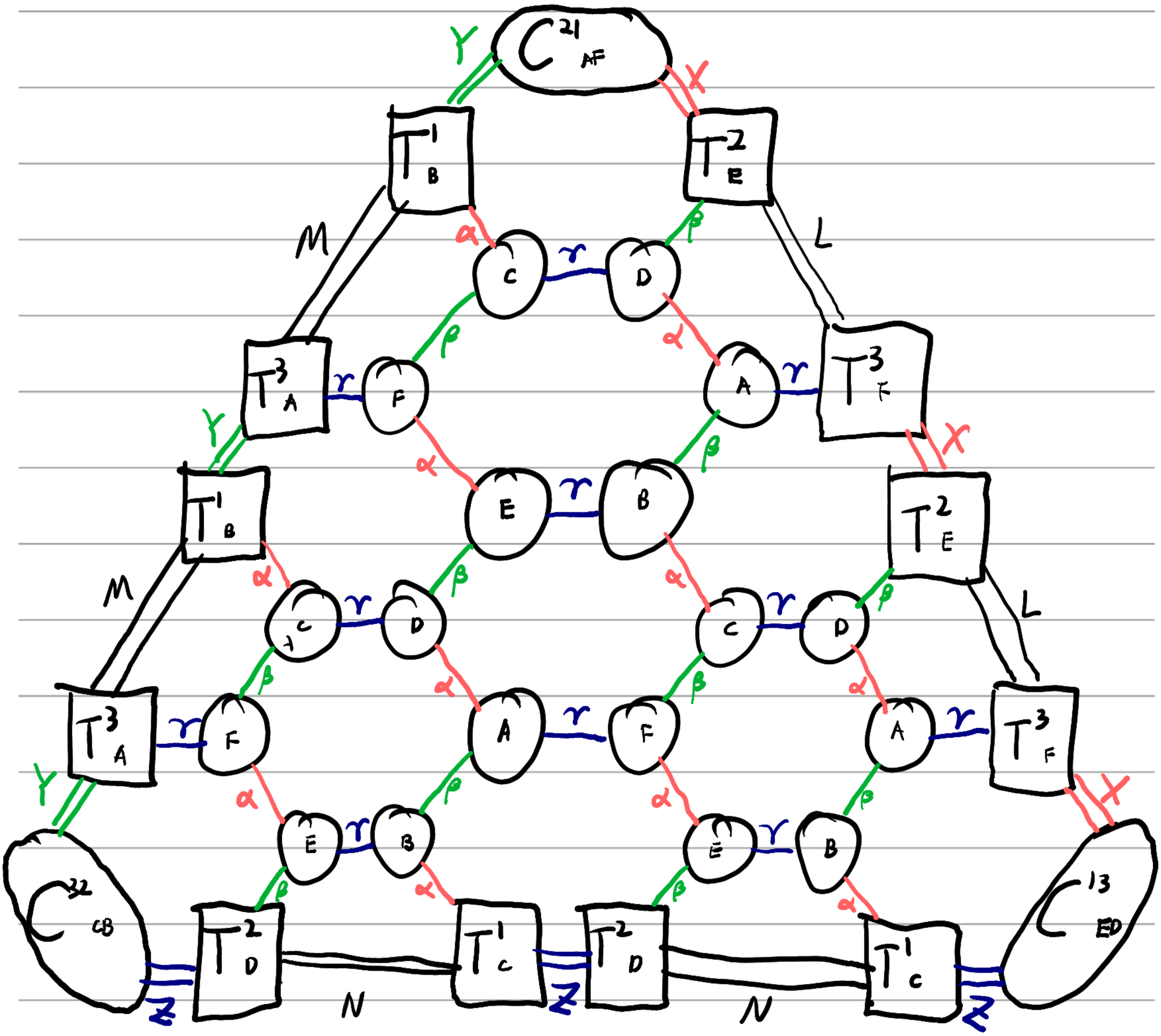


C^{13}: AB → CF → ED [XZ]

$T^3 T^1$ [M, Y, γα]



3 center $\begin{matrix} & E & B \\ D & & & C \\ & A & F \end{matrix}$



track → discarded work
(biggest truncation)
(the ε related to x!)

good

guess makes
beginning SVD
more stable!